DERIVATIVES AND SHAPE QUIZ

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Name: Solutions

Use the function

$$f(x) = x^4 - 2x^3$$

to answer the following questions.

1. Find the intervals where the function is increasing and decreasing. Write NONE if not applicable.

Increasing: $(3/2, \infty)$

Decreasing: $(-\infty, 0) \cup (0, 3/2)$

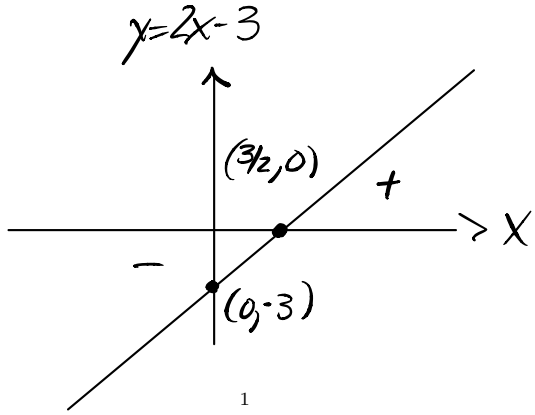
$f'(x) = 4x^3 - 6x^2 = 2x^2(2x-3) = 0 \iff x=0$ or $x = 3/2$

The sign of $f'(x)$ depends only on $2x-3$ since $2x^2 \geq 0$.

2. State the local maximum and local minimum value(s). Write NONE if not applicable.

Local maximum value(s): None

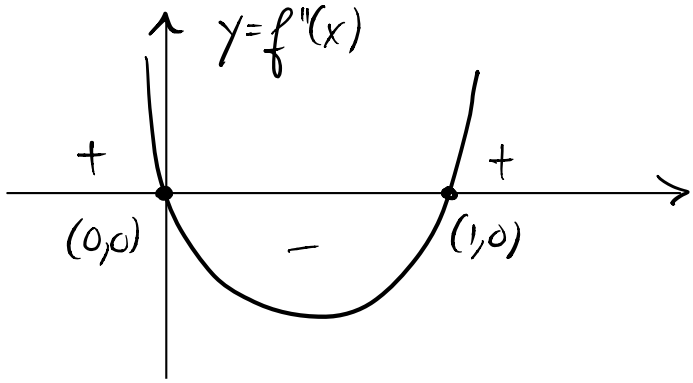
Local minimum value(s): $(3/2, -27/8)$



3. Find the intervals on which the function is concave up and concave down. State the inflection points. Write NONE if not applicable.

Concave Up: $(-\infty, 0) \cup (1, \infty)$

Concave Down: $(0, 1)$

Inflection Points: $(0,0)$ and $(1,-1)$

$$f''(x) = 12x^2 - 12x = 12x(x-1)$$